

Predicate Logic

COMP2711 Discrete Mathematical Tools

Rosen, Sections 1.4–1.5

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SECTION 1

Motivation

Propositions and predicates

Propositional logic

P_A : "Alice submitted,"

P_B : "Bob submitted,"

P_C : "Carol submitted."

Propositions and predicates

Propositional logic

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To say that every student submitted:

$$P_A \wedge P_B \wedge P_C \wedge \dots$$

We need one proposition per student.

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Predicate logic

Domain: all students in the class.

Let $S(x)$ mean

“ x submitted the assignment.”

Then one formula says the whole claim:

$$\boxed{\forall x S(x)}.$$

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Propositional logic

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Elements of predicate logic

Domain: $\mathbb{Z}$

$\exists x \text{ Even}(x)$

“There is an even integer.”

Elements of predicate logic

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x : an object name

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Domain: $\mathbb{Z}$

$\exists x \text{ Even}(x)$

"There is an even integer."

variables

x : an object name

predicate

$\text{Even}(x)$: a property of x

Elements of predicate logic



Domains

$\mathbb{Z}$ = all integers = $\{\dots, -2, -1, 0, 1, 2, \dots\}$

$\mathbb{R}$ = all real numbers.

Consider the open formula

$$x^2 \geq x.$$

Unless stated otherwise, we assume domains are nonempty.

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The symbols did not change. The domain did.

Unless stated otherwise, we assume domains are nonempty.

SECTION 2

Quantifiers

Universal and existential quantifiers

$$\forall x P(x)$$

$$\exists x P(x)$$

Universal and existential quantifiers

$$\forall x P(x)$$

Meaning

For all x in the domain, $P(x)$ is true.

$$\exists x P(x)$$

Meaning

There is some x in the domain such that $P(x)$ is true.

Universal and existential quantifiers

$$\forall x P(x)$$

Meaning

For all x in the domain, $P(x)$ is true.

To prove it

Prove or check $P(x)$ for all x .

$$\exists x P(x)$$

Meaning

There is some x in the domain such that $P(x)$ is true.

To prove it

Just find one x for which $P(x)$ is true.

Universal and existential quantifiers

$$\forall x P(x)$$

Meaning

For all x in the domain, $P(x)$ is true.

To prove it

Prove or check $P(x)$ for all x .

To disprove it

Find one counterexample x for which $P(x)$ is false.

$$\exists x P(x)$$

Meaning

There is some x in the domain such that $P(x)$ is true.

To prove it

Just find one x for which $P(x)$ is true.

To disprove it

Show $P(x)$ is false for all x .

Free and bound variables

Set the domain to $\mathbb{Z}$, the set of all integers.

Once the domain is fixed as $\mathbb{Z}$, we often shorten this to $\exists x \text{ Prime}(x)$.

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Open formula

$\text{Prime}(x)$

“ x is prime.”

Its truth depends on x , so x is **free**.

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$\underbrace{\exists x \in \mathbb{Z}}_{\text{quantifier + domain}} \underbrace{\text{Prime}(x)}_{\text{scope}}$

“There exists an integer x such that x is prime.”

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When we **bind a variable with a quantifier within a domain**, it is no longer free.
When a formula has no free variables, it is a **proposition** that is either true or false.

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Proposition

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Nested quantifiers

Start with the sentence:

Every integer greater than 1 has a prime divisor.

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$\forall n$

choose every integer

Nested quantifiers

Start with the sentence:

Every integer greater than 1 has a prime divisor.

$$\forall n(n > 1 \rightarrow \dots)$$

restrict the requirement to integers greater than 1

Nested quantifiers

Start with the sentence:

Every integer greater than 1 has a prime divisor.

$$\forall n(n > 1 \rightarrow \exists p(\dots))$$

ask for a prime divisor

Nested quantifiers

Start with the sentence:

Every integer greater than 1 has a prime divisor.

$$\forall n \left(n > 1 \rightarrow \exists p (\text{Prime}(p) \wedge p \mid n) \right)$$

Read aloud: "For every integer n , if $n > 1$, then there exists a prime p that divides n ."

Quantifier negation

$$\neg \forall x P(x) \equiv \exists x \neg P(x),$$

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No object works $\iff$ every object fails.

Negation example

Let $H(n) := \exists p(\text{Prime}(p) \wedge p \mid n)$.

Read $H(n)$: “ n has a prime divisor.”

Task: Negate $\forall n(n > 1 \rightarrow H(n))$.

In words: “Every integer greater than 1 has a prime divisor.”

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expand H ; negate $\exists$

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In English: **some integer greater than 1 has no prime divisor.** This is false.

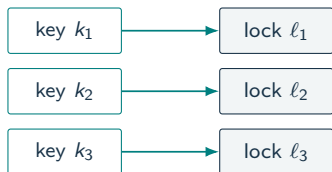
SECTION 3

Quantifier order

Quantifier-order fallacy

Let $\text{Opens}(k, \ell)$ mean “key k opens lock ℓ .”

Every lock has some key

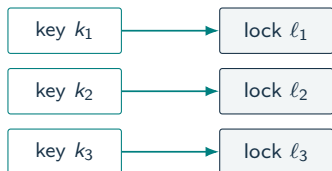


$$\forall \ell \exists k \text{ Opens}(k, \ell)$$

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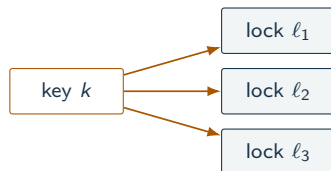
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$$\forall \ell \exists k \text{ Opens}(k, \ell)$$



One key opens every lock



$$\exists k \forall \ell \text{ Opens}(k, \ell)$$

Reordering quantifiers

Same type: swapping preserves meaning

$$\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$$

“ P holds for every pair.”

$$\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$$

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“P holds for some pair.”

Mixed types: one direction is always valid

$$\exists y \forall x R(x, y) \implies \forall x \exists y R(x, y)$$

“One y works for every x” implies “for every x, some y works.”

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“One y works for every x” implies “for every x, some y works.”

The reverse direction is generally false

$$\forall x \exists y R(x, y) \not\Rightarrow \exists y \forall x R(x, y)$$

“Each x may have its own y” does not imply “one y works for every x.”

SECTION 4

Translation

Translation method

1. **State the domain.**
2. **Define predicates** and their argument order.
3. **Determine the choices:** what is selected first?
4. **Write the formula:** apply restrictions correctly.
5. **Read back** and test a small example.

Translation is not word replacement. It is reconstruction of the sentence's logic.

Existence and uniqueness

Translate: **Every even integer has exactly one integer half.**

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Existence for a fixed n

$$\exists k (n = 2k)$$

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Uniqueness after choosing k

$$\forall j (n = 2j \rightarrow j = k)$$

Existence and uniqueness

Translate: **Every even integer has exactly one integer half.**

$$\forall n (\text{Even}(n) \rightarrow \exists k (n = 2k \wedge \forall j (n = 2j \rightarrow j = k)))$$

Existence and uniqueness

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One such value exists, and any value that works must be that same value. The notation $\exists!k$ abbreviates this pattern.

Writing “exactly one”

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Together:

$$\exists y \left(C(y) \wedge \forall z (C(z) \rightarrow z = y) \right).$$

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Together:

$$\exists y \left(C(y) \wedge \forall z (C(z) \rightarrow z = y) \right).$$

For “every integer has exactly one additive inverse,” use $C(y) : x + y = 0$:

$$\forall x \in \mathbb{Z} \exists y \in \mathbb{Z} (x + y = 0 \wedge \forall z \in \mathbb{Z} (x + z = 0 \rightarrow z = y)).$$

Mathematical writing

In papers and textbooks, writers rarely stay at either extreme. Consider Goldbach's conjecture, with all variables ranging over $\mathbb{Z}$.

Fully verbal

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Every even integer greater than two is the sum of two prime numbers.

Fully symbolic

$$\forall n \left((\text{Even}(n) \wedge n > 2) \rightarrow \exists p \exists q (\text{Prime}(p) \wedge \text{Prime}(q) \wedge n = p + q) \right).$$

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Typical mathematical prose

Every even integer $n > 2$ can be written as $n = p + q$ for primes p and q .

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Typical mathematical prose

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The usual register uses **words for meaning** and **notation for structure**.

Translation practice

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$$\exists p \left(\text{Prime}(p) \wedge \forall n (\text{Even}(n) \rightarrow p \mid n) \right).$$

Translation practice

Domain: the integers.

Some prime divides every even integer.

$$\exists p \left(\text{Prime}(p) \wedge \forall n (\text{Even}(n) \rightarrow p \mid n) \right).$$

Choose $p = 2$, then check an arbitrary even integer n .

SECTION 5

Equivalence and normal form

Valid distribution laws

Universal quantifier with “and”

$$\forall x (P(x) \wedge Q(x)) \equiv (\forall x P(x)) \wedge (\forall x Q(x))$$

“Every object has both properties P and Q ” means “every object has P , and every object has Q .”

Valid distribution laws

Universal quantifier with “and”

$$\forall x (P(x) \wedge Q(x)) \equiv (\forall x P(x)) \wedge (\forall x Q(x))$$

“Every object has both properties P and Q” means “every object has P, and every object has Q.”

Existential quantifier with “or”

$$\exists x (P(x) \vee Q(x)) \equiv (\exists x P(x)) \vee (\exists x Q(x))$$

“Some object has property P or Q” means “some object has P, or some object has Q.”

Universal quantification and disjunction

Always valid

$$(\forall x P(x)) \vee (\forall x Q(x)) \implies \forall x (P(x) \vee Q(x))$$

If one property holds for every object, then every object has at least one of the two properties.

Universal quantification and disjunction

The converse can fail

Domain: all integers.

Even integers: $\dots, -2, 0, 2, 4, \dots$

Odd integers: $\dots, -3, -1, 1, 3, \dots$

True in this model

$$\forall x (\text{Even}(x) \vee \text{Odd}(x))$$

“Every integer is even or odd.”



False in this model

$$(\forall x \text{ Even}(x)) \vee (\forall x \text{ Odd}(x))$$

“Every integer is even, or every integer is odd.”

Existential quantification and conjunction

Always valid

$$\exists x (P(x) \wedge Q(x)) \implies (\exists x P(x)) \wedge (\exists x Q(x))$$

If one object has both properties, then each separate existence claim is true.

Existential quantification and conjunction

The converse can fail

Domain: all integers.

Even integers: $\dots, -2, 0, 2, 4, \dots$

Odd integers: $\dots, -3, -1, 1, 3, \dots$

True in this model

$$(\exists x \text{ Even}(x)) \wedge (\exists x \text{ Odd}(x))$$

“Some integer is even; some integer is odd.”



False in this model

$$\exists x (\text{Even}(x) \wedge \text{Odd}(x))$$

“Some integer is both even and odd.”

Prenex normal form

Example already in PNF

$$\underbrace{\forall x \exists y}_{\text{all quantifiers first}} \underbrace{(P(x) \vee R(x, y))}_{\text{no quantifiers inside}} .$$

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The first part is the **quantifier prefix**; the remaining formula is the **matrix**. To put another formula into this shape:

1. Rename bound variables to distinct names.
2. Eliminate $\rightarrow$ and $\leftrightarrow$.
3. Move negations inward, switching crossed quantifiers.
4. Pull quantifiers outward.

Prenex normal form

Example already in PNF

$$\underbrace{\forall x \exists y}_{\text{all quantifiers first}} \quad \underbrace{(P(x) \vee R(x, y))}_{\text{no quantifiers inside}}.$$

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Pull a quantifier past another part only when its variable is not free there. Do one justified transformation per line.

PNF conversion

Convert

$$\exists z \left(\exists y R(y, z) \rightarrow \forall y Q(y, z) \right)$$

to prenex normal form.

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PNF conversion steps

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remove the arrow

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move negation inward

$$\equiv \boxed{\exists z \forall y \forall x (\neg R(y, z) \vee Q(x, z))}$$

pull quantifiers outward.

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$$\equiv \boxed{\exists z \forall y \forall x (\neg R(y, z) \vee Q(x, z))}$$

pull quantifiers outward.

The remaining formula $\neg R(y, z) \vee Q(x, z)$ has no quantifiers.

Summary

1. Predicates describe properties and relations over a stated domain.
 2. Quantifiers bind variables and determine the evidence a claim needs.
 3. Quantifier order determines whether one choice may depend on another; scope controls meaning.
 4. Negation, counterexamples, and PNF transformations must preserve that structure.
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Next lecture

Inference and proof: deriving conclusions from quantified premises.